NovaTech Supply Chain Control Tower- (Power BI Dashboard Project)

1. Project Overview

This project focuses on analyzing supply chain performance using Power BI. The dashboard provides insights into delivery performance, order fulfilment, stock availability, supplier performance and inventory management.

It allows users to interactively analyze supply chain data using filters for Region, Category, Supplier and Month.

2. Project Objectives

- Monitor On-Time In-Full (OTIF) delivery performance.
- Analyze order fulfilment and fill rate.
- Identify stockout issues across product categories.
- Evaluate supplier and regional performance.
- Monitor inventory value, inventory turns and days inventory.
- Identify areas requiring improvement in supply chain operations.

3. Tools and Technologies

Power BI, Power Query, DAX, Data Visualization

4. Key Performance Indicators (KPIs)

KPI	Value
OTIF %	67.61%
Fill Rate %	98.86%
Stockout %	19.78%
Inventory Turns	87.39%
Days Inventory	417.66

5. Key Dashboard Features

- Interactive filters for Region, Category, Supplier and Month.
- OTIF performance by region.
- OTIF performance by Supplier.
- Stockout percentage by Category.
- Regional supply chain performance matrix.
- Inventory Value by Category.
- Calculated OTIF vs Supplied OTIF comparison.
- KPI cards with trend indicators.

6. Key Insights

➢ Regional Performance

- Central recorded the highest OTIF at 76.38%.
- North recorded the lowest OTIF at 54.78%.
- North had the longest average lead time of 22.35 days.
- Central had the shortest average lead time of 19.09 days.

➢ Order Fulfilment

- Overall fill rate was 98.86%.
- Despite the high fill rate, overall OTIF was 67.61%, indicating a gap between order fulfilment and on-time delivery performance.

➢ Inventory and Stockouts

- Critical Components recorded the highest stockout percentage among categories.
- Overall stockout rate was 19.78%.
- Days Inventory was 417.66.

➢ Supplier Performance

- Crest Global recorded the lowest OTIF among the displayed suppliers.
- Supplier-level analysis helps identify differences in delivery performance.

7. Business Recommendations

- Investigate delivery delays in the North region.
- Review supplier performance, particularly Crest Global.
- Improve stock availability for Critical Components.
- Monitor inventory levels to improve inventory efficiency.
- Investigate the reasons behind the gap between fill rate and OTIF.

8. Conclusion

The Power BI dashboard provides an interactive view of supply chain operations. It helps identify regional performance gaps, supplier delivery issues and inventory-related challenges.

The analysis can support better operational monitoring and data-driven decision-making.

9. Project Files

Power BI dashboard (.pbix), Dataset, Dashboard screenshots